

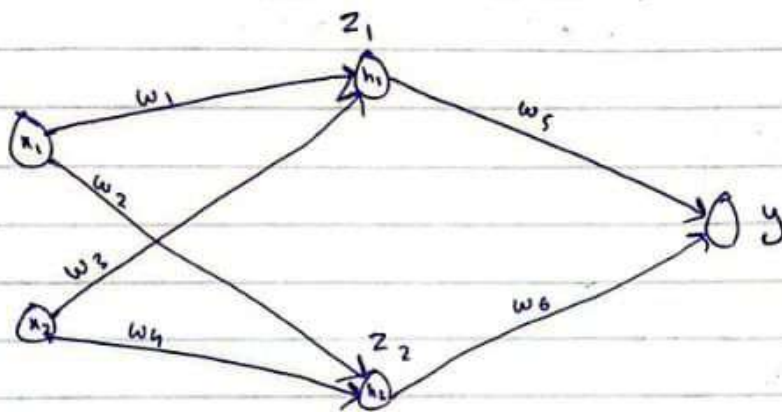
Initial Setup:

$$x = \begin{bmatrix} x_1 & 0.5 \\ x_2 & 0.1 \end{bmatrix}$$

$$y = 1$$

$$W_1 = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}$$

$$W_2 = [0.5, 0.6], \quad b_2 = [0.0]$$



Weighted Sum:

$$\rightarrow z_1 = W_1 \cdot x + b_1$$

$$z_1 = x_1 \cdot w_1 + x_2 \cdot w_3$$

$$z_2 = x_1 \cdot w_2 + x_2 \cdot w_4$$

$$z_{1,1} = 0.1 \times 0.5 + 0.2 \times 0.1$$

$$z_{1,1} = 0.05 + 0.02$$

$$z_{1,1} = 0.07$$

$$z_2 = x_1 \cdot w_2 + x_2 \cdot w_4$$

$$= 0.3 \times 0.5 + 0.1 \times 0.6$$

$$= 0.15 + 0.06$$

$$(z_2 = 0.19)$$

$$z_1 = \begin{bmatrix} 0.07 \\ 0.19 \end{bmatrix}$$

$$h_1 = z_1 \times w_5$$

$$\rightarrow a_1 = \sigma(z_1)$$

$$a_{1,1} = \sigma(0.07) = \frac{1}{1 + e^{-0.07}} = \frac{1}{1 + 0.9324} = 0.5175$$

$$a_{1,2} = \sigma(0.19) = \frac{1}{1 + e^{-0.19}} = \frac{1}{1 + 0.8270} = 0.5474$$

$$a_1 = \begin{bmatrix} 0.5175 \\ 0.5474 \end{bmatrix}$$

$$\rightarrow z_2 = W_2 \cdot a_1 + b_2$$

$$z_2 = (0.5)(0.5175) + (0.6)(0.5474) + 0.0$$

$$z_2 = 0.2588 + 0.3284 = 0.5872$$

$$\rightarrow \hat{y} = \sigma(z_2)$$

$$\hat{y} = \sigma(0.5872) = \frac{1}{1 + e^{-0.5872}} = \frac{1}{1 + 0.5560} = 0.6427$$

$$\rightarrow \text{Loss}$$

$$L = -\log(\hat{y}) = -\log(0.6427) = 0.4420$$